

where

$$B_{n-1} = \begin{bmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_{n-1} \end{bmatrix}. \quad (14)$$

From (2) and (6) we compute the estimate of $\bar{x}_n$, given $\bar{z}_{n-1}$ (but not $\bar{\phi}_n$), to be

$$\begin{aligned} \hat{x}_n &= E[\bar{x}_n \bar{y}_{n-1}^T] B_{n-1} \{ E[\bar{z}_{n-1} \bar{z}_{n-1}^T] \}^{-1} \bar{z}_{n-1} \\ &= E[\bar{x}_n \bar{y}_{n-1}^T] \{ E[\bar{y}_{n-1} \bar{y}_{n-1}^T] \}^{-1} \{ E[\bar{y}_{n-1} \bar{z}_{n-1}^T] \} \\ &\quad \cdot B_{n-1} E[\bar{z}_{n-1} \bar{z}_{n-1}^T]^{-1} \bar{z}_{n-1} \\ &= \{ \text{estimator of the form (6)} \} \\ &\quad \cdot \{ \text{LUMV estimate of } \bar{y}_{n-1} \} \\ &= [0, 0, \dots, R_{n,n-1}] \bar{y}_{n-1}^* \\ &= R_{n,n-1} \bar{x}_{n-1}^*. \end{aligned} \quad (15)$$

Similarly, we calculate the error variance

$$\hat{\Lambda}_n = E[(\bar{x}_n - \hat{x}_n)(\bar{x}_n - \hat{x}_n)^T]$$

as given by (11), noting from (2) and (3) that

$$E[\bar{y}_{n-1}^* \bar{y}_{n-1}^{*T}] = [\Lambda_{y,n-1} - \Lambda_{y,n-1}^*].$$

To complete the theorem, we define the conditional random variable $\Delta \hat{x}_n = (\bar{x}_n - \hat{x}_n)$, and introduce the data at time t_n in the form $\Delta \bar{\phi}_n = \bar{\phi}_n - A_n \hat{x}_n = A_n \Delta \bar{x}_n$. Since

$$E[\Delta \bar{x}_n \bar{z}_{n-1}^T] = 0 \quad \text{and} \quad E[\Delta \bar{\phi}_n \bar{z}_{n-1}^T] = 0,$$

it follows (Theorem 1) that the LUMV estimate $\Delta \hat{x}_n^*$, given $\bar{z}_{n-1}$ and $\Delta \bar{\phi}_n$, only depends upon $\Delta \bar{\phi}_n$. Thus

$$\Delta \hat{x}_n^* = E[\Delta \bar{x}_n \Delta \bar{\phi}_n^T] (E[\Delta \bar{\phi}_n \Delta \bar{\phi}_n^T])^{-1} \Delta \bar{\phi}_n \quad (16)$$

Equations (9) and (12) follow from (16)

It is interesting to note that the forms of the estimation and covariance equations change when one seeks to obtain the continuous estimator by letting the time between data samples go to zero. In particular, in Pfeiffer [5] it is shown that the difference equation for the discrete estimate converges to a linear differential equation which contains the data derivative but not the data itself. The derivation of the continuous case does not depend upon introducing the notion of continuous white noise, indeed, (4) and (5) make no sense if $P(t, s)$ contains a Dirac delta function.

CARL PFEIFFER
Space Guidance Theory Group
Jet Propulsion Lab.
Pasadena, Calif.

REFERENCES

- [1] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Engrg.*, pp. 35-45, March 1960.
- [2] R. E. Kalman, and R. S. Bucy, "New results in linear filtering and prediction theory," *J. Basic Engrg.*, pp. 95-108, March 1961.
- [3] R. E. Kalman, "New methods and results in linear prediction and filtering theory," Research Institute for Advanced Study, Baltimore, Md., Tech. Rept. 61-1, June 1961.
- [4] C. G. Pfeiffer, "Sequential estimation of correlated stochastic variables," Jet Propulsion Lab., Pasadena, Calif., Tech. Rept. 32-445, July 1963.
- [5] C. G. Pfeiffer, "Continuous estimation of sequentially correlated random variables," Jet Propulsion Lab., Pasadena, Calif., Tech. Rept. 32-524, October 1963.
- [6] J. L. Doob, *Stochastic Processes*. New York: Wiley, 1953.

On a New Measure of Interaction for Multivariable Process Control

INTRODUCTION

This correspondence describes an interaction measure for multivariable control developed to overcome theoretical and practical deficiencies of the matrix representation as a tool for design of systems. A linearized, time-invariant, multivariable process described by a square gain matrix input-output relation is assumed. Simple extensions of the measure and its use to handle dynamics, nonlinearity, and nonsquare matrices exist. It is usually better, however, to use the unmodified measure to determine the initial design structure of control allowing dynamic behavior and nonlinearity to determine later details.

The control problem discussed is shown in Fig. 1 with m_j and c_i reconnected where the process is described by the matrix-vector relation $\Delta c = \Phi \Delta m$ with Φ a process gain matrix whose elements are indicated as ϕ_{ij} . However, the matrix by itself is a poor tool for classifying control properties of the system. It is highly dependent on scaling (see Property 2) and choice of units. For instance, the two matrices

$$\begin{pmatrix} 10 & 10 \\ 0 & 10 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 10 & 10 \\ 9 & 10 \end{pmatrix}$$

having radically different control characteristics, can be scaled by the same change of units to apparently similar matrices

$$\begin{pmatrix} 1 & 10 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 10 \\ .09 & 1 \end{pmatrix}.$$

Scaling becomes more confusing with large systems. Conventional properties of a matrix, including norms, the determinant and the eigenvalues also depend on scaling or on the ordering of rows and columns in the matrix. And the properties that are most related to control such as the eigenvalues are also most dependent on variable scaling and ordering, making them unsuitable as measures of process structure.

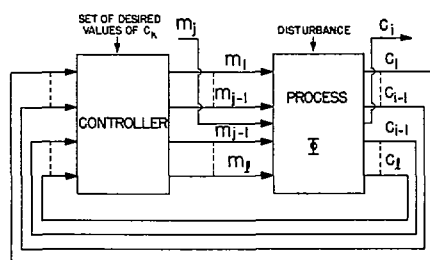


Fig. 1. Controlled multivariable system with an input-output pair disconnected from control.

The term interaction arose because industry has often found it desirable to control the multivariable process as if it were made up of isolated single variable processes. The resulting loops "interact." An interaction measure attempts to answer the question:

How is the measured transfer function between a given manipulated variable m_j and a given controlled variable c_i affected by the complete control of all other controlled variables, as in Fig. 1?

DEFINITION

The measure taken to answer this question is the ratio of two gains representing first the process gain in an isolated loop and, second, the apparent process gain in that same loop when all other control loops are closed. In the first case, the gain between c_i and m_j is $[(\Delta c_i / \Delta m_j) | \Delta m_k = 0 \text{ when } k \neq j] = \phi_{ij}$, the open loop gain. In the second case, the steady-state gain is

$$[(\Delta c_i / \Delta m_j) | \Delta c_k = 0 \text{ when } 0 \neq k \neq i] = 1 / \phi_{ji}^{-1},$$

where ϕ_{ji}^{-1} is an element in the inverse matrix. Notice that the abstraction represented by these gains does not depend on the structure chosen for the controller except that it be effective in enforcing steady-state values of c equal to the desired values. The ratio of these gains defines an array M with elements:

$$\mu_{ij} \triangleq \phi_{ij} \phi_{ji}^{-1}.$$

This definition is related in statement and purpose to the "condition number" suggested by von Neumann and Goldstine [2]. Other quite different measures have been proposed [3]-[5]. The author was recently referred to related work of Fiedler and Ptak [6].

PROPERTIES OF THE MEASURE

The following properties are easily shown to be true:

- 1) Any row or column sums to one.
- 2) The measure of a matrix is invariant under scaling. (Scaling of a matrix corresponds to the multiplication of the matrix by two general diagonal matrices D' and D so that the scaled matrix becomes $\Phi' = D' \Phi D$.)
- 3) The only effect of altering the order of rows or columns in Φ is to introduce the same alteration of order in M .
- 4) Measures much larger than one imply a "nearly" singular gain matrix.
- 5) The submeasure of an effectively isolated subprocess is the same as the measure of the subprocess. All other elements of rows and columns common to the submeasure are zero.
- 6) The measure shows up in calculations of the changes introduced in a control system caused by changes in process parameters (and nonlinearity); for instance, the relation

$$d\mu_{ij} = \mu_{ij}(d\phi_{ij}/\phi_{ij} + d\phi_{ji}^{-1}/\phi_{ji}^{-1}).$$

The measure is an approximate measure of its own sensitivity.

Properties 7 and 8 relate to interaction induced dynamic behavior. Both properties are generally true and hold rigorously if integral control action dominates all other process and control dynamics. This is not a

Manuscript received October 16, 1964; revised May 26, and October 8, 1965.

trivial case since it corresponds to "loose" or "conservative" control and sets the tenor of practical control.

- 7) The transfer function between c_i and m_j measured as in Fig. 1 with all other loops closed will be nonminimum phase or unstable if μ_{ij} is negative.
- 8) A two-by-two process controlled by two negative feedback controllers set loosely as before in a minimum loop system must be stable if controllers are assigned to variable pairs with positive measure. The same system but based on negative measure will be stable only if one loop is in positive feedback and then only for certain ratios of loop gains with the negative feedback loop closed. Similar care is required in more complex systems. Detailed generalizations can be proven for three-by-three processes and, therefore, for all systems in which two-by-two or three-by-three subprocesses dominate.

EXAMPLES AND DISCUSSION

Figure 2 shows five examples of gain matrices of two-by-two multivariable processes along with the measure calculated from them. In Fig. 2, (a), (b), and (c) are any nonzero numbers and δ is a nonzero number of absolute value much less than one. Figure 2(a) and (b) both show processes with no interaction. The process of Fig. 2(c) gives a measure which also appears to show noninteraction as would any triangular matrix or matrix obtained by permuting rows or columns of a triangular matrix. The scaling

$$D = \begin{pmatrix} \alpha & 0 \\ 0 & 1/\alpha \end{pmatrix}, \quad D' = \begin{pmatrix} 1/\alpha & 0 \\ 0 & \alpha \end{pmatrix}$$

scales Figs. 2(c) to 2(a) in the limit as α approaches zero. Any two loop control system applied stably to any dynamic process whose gain matrix is that of Fig. 2(a) will also be stable when applied to Fig. 2(c). Decoupling for this class of effectively non-interacting process amounts to simple feed-forward and is also never destabilizing. Stability is unchanged because in each case no new loop is introduced by the interaction.

The process of Fig. 2(d) shows almost identical effects of each manipulated variable on each controlled variable. For this reason, independent control of each controlled variable is difficult to achieve practically. Near singularity is shown by Property 4. Existence of negative measure is a necessary characteristic of nearly singular systems by Properties 4 and 1. Other problems are made clear from the measure by Properties 6, 7, and 8.

In contrast to Fig. 2(d), Fig. 2(e) shows a process in which each manipulated variable has large but differing effects on each controlled variable. Control actions on independent m_i do not cancel one another, and the system is more easily and insensitively controlled with or without decoupling. The character of the matrix is reflected in an interaction measure having positive values less than one.

$$\begin{aligned} \text{(a)} \quad \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} &\longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \text{(b)} \quad \begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix} &\longrightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \text{(c)} \quad \begin{bmatrix} a & 0 \\ c & b \end{bmatrix} &\longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \text{(d)} \quad \begin{bmatrix} a & a \\ a & \alpha(1+\delta) \end{bmatrix} &\longrightarrow \begin{bmatrix} \frac{1+\delta}{\delta} & -\frac{1}{\delta} \\ -\frac{1}{\delta} & \frac{1+\delta}{\delta} \end{bmatrix} \\ \text{(e)} \quad \begin{bmatrix} a & a \\ -a & a \end{bmatrix} &\longrightarrow \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} \end{aligned}$$

Fig. 2. Table of gain matrices with their interaction measure tables for: (a) and (b) noninteracting processes, (c) an interacting process whose interaction is free of feedback paths, (d) a highly interacting process with considerable control difficulty, and (e) a highly interacting process, whose interaction is easily decoupled.

CONCLUSION

The basic character of these types of interaction carries over into n -by- n processes. The measure can serve as a design tool to select preferred processes and to specify the control structure once a process is selected. This control structure is specified by a one-to-one pairing of the controlled and manipulated variables as a basis for control. Each pair may be closed in a single loop in a minimum loop system or more refined decoupling may be used. In any case, the procedure amounts to picking a preferred principal diagonal to the matrix. The examples and properties suggest that the measure corresponding to the paired variables be positive and as close to one as possible. Numbers negative or much larger than one are to be avoided and large negative numbers are particularly undesirable.

This design procedure is hardly complete but it is simple to use, and it usually gives a unique design even when stated in such qualitative terms. It is supported by Property 8 as well as the intuitive argument and has correlated perfectly with standard designs of actual industrial processes. The author hopes to be able to free the supporting experimental data for publication at some time in the future.

EDGAR H. BRISTOL
Foxboro Company
Foxboro, Mass.

REFERENCES

- [1] F. B. Hildebrand, *Methods of Applied Mathematics*, Englewood Cliffs, N. J.: Prentice-Hall, 1952.
- [2] J. von Neumann and H. H. Goldstine, "Numerical inverting of matrices of high order," *Proc. Amer. Math. Soc.*, vol. 2, pp. 188-202, 1951.
- [3] M. D. Mesarovic, "A measure of interaction and its application to control problems," Systems Research Center, Case Institute of Technology, Cleveland, Ohio, Rept. A-6-60.
- [4] R. Brockert, "The control of linear multivariable systems," Ph.D. thesis, Case Institute of Technology, Cleveland, Ohio, 1962.
- [5] P. C. Chiu and C. R. Webb, "Analog computer study of a multivariable control system," *Control*, vol. 5, no. 49, pp. 77-80.
- [6] M. Fielder and V. Ptak, "On matrices with non-positive off-diagonal elements and positive principal minors," *Chechoslovakii Matematicheski Zhurnal*, vol. 12, pp. 382-400, 1962.

Dead Beat Response of Higher-Order Servo Systems by Single Switching Operation

Smith suggested a method [1] for obtaining dead beat response of lightly damped second-order servo systems to step signal inputs. In this method, the input command has to be suitably controlled, which can be effected by single switching operation [2], [3]. By such control of the input signal it is possible to obtain dead beat response of higher-order systems provided the systems have suitable transfer functions. The purpose of the present correspondence is to investigate these transfer functions of higher-order systems.

Consider a closed loop transfer function of an n th-order system as

$$H(S) = \frac{A(S)}{B(S)} = \frac{b_0}{b_n S^n + b_{n-1} S^{n-1} + \dots + b_1 S + b_0} \quad (1)$$

where the coefficients b are all real constants and n are positive integers. The transform of the output response $C(S)$, for a unit step excitation function and for zero initial conditions is

$$C(S) = \frac{1}{S} \frac{A(S)}{B(S)} \quad (2)$$

We shall assume that the roots of the equation $B(S) = 0$ are all distinct. Let $S_1, S_2, \dots, S_n$ be the n distinct roots. Then $C(S)$ can be expanded in the following form

$$C(S) = \frac{1}{S} + \sum_{i=1}^n \frac{A_i}{S + S_i} \quad (3)$$

where the numerators of the partial fractions in (3) are constants. The output response $C(t)$ is given by

$$C(t) = 1 + \sum_{i=1}^n A_i \exp(-S_i t) \quad (4)$$

The expression for the q th derivative of the output can be written as

$$\frac{d^q C}{dt^q} = (-1)^q \sum_{i=1}^n S_i^q A_i \exp(-S_i t) \quad (5)$$

Since the initial conditions are zero, it follows from (5) that the following relation is satisfied

$$\sum_{i=1}^n S_i^q A_i = 0 \quad (6)$$

for all values of q lying between 1 to $n-1$.

Now, if at any finite time, all the derivatives for the previous values of q are to attain zero values simultaneously, all exponential terms in (5) must have the same value at that time. Since the roots are distinct, this can occur when the real parts of all the roots are equal, that is, when the roots lie on a line parallel to the imaginary axis of the complex plane. However, for obtaining dead beat response by the single switching operation it is required that all the previous derivatives should attain zero values by the time the first derivative